SENTINEL
(SEWER ENVIRONMENTAL NETWORK FOR THREAT INTELLIGENCE AND ENVIRONMENTAL LEARNING)

TEAM **BATCH - B2**

KARTHICK RAJA P 2127220701062
KARUNYA R 2127220701065
MOHAMMED RAEEF 2127220701091

MENTOR: Dr.G.Ayappan, Assistant Professor ECE

ABSTRACT

SENTINEL (Sewer Environmental Network for Threat Intelligence and Environmental Learning) is an IoT-based monitoring system designed for real-time sewer environment surveillance. The system uses MQ-7, MQ-135, and DHT11 sensors to monitor hazardous gases, temperature, and humidity, while an ESP32-S3 camera captures images for pest detection. A MobileNetV3 model integrated with CBAM performs insect classification, and the results are transmitted via MQTT to a web-based dashboard for centralized monitoring. By combining environmental sensing, AI-powered pest detection, and real-time visualization, SENTINEL provides a scalable and cost-effective solution for improving sewer safety and maintenance.

HARDWARE & SOFTWARE REQUIREMENTS

HARDWARE

- ESP32 Microcontroller
- ESP32-S3 Camera Module
- MQ-7 Gas Sensor
- MQ-135 Gas Sensor
- DHT11 Temperature & Humidity Sensor
- Wi-Fi

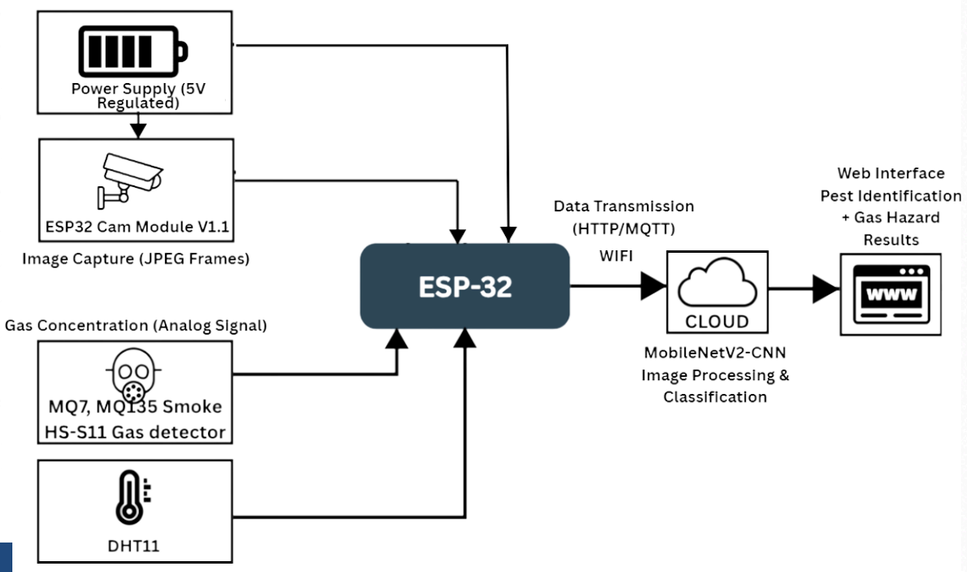
SOFTWARE

- Arduino IDE
- Python
- MQTT Protocol
- Firebase / Cloud Database
- MobileNetV3
- CBAM Attention Module
- Web Dashboard

PROPOSED SOLUTION

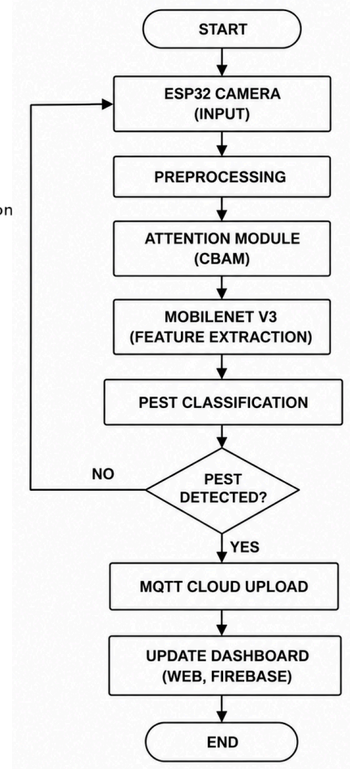
The SENTINEL system is an IoT-based sewer monitoring solution that integrates environmental sensing and AI-powered pest detection. An ESP32 collects data from MQ-7, MQ-135, and DHT11 sensors, while an ESP32-S3 camera captures images for insect classification using a MobileNetV3-CBAM model. The processed data is transmitted via MQTT to a cloud platform and visualized through a web dashboard, enabling real-time monitoring, intelligent alerts, and improved environmental safety.

BLOCK DIAGRAM



FLOWCHART (MQ7, MQ135, DHT 11)

FLOWCHART ESP32-CAM



PROJECT MODULES

Environmental Monitoring Module
Monitors hazardous gases, temperature, and humidity using MQ-7, MQ-135, and DHT11 sensors.

Pest Detection Module
Captures sewer images and classifies pests using MobileNetV3 integrated with CBAM.

Edge Processing Module
Processes sensor data locally and performs threshold-based hazard detection.

MQTT Communication Module
Transmits environmental and pest detection data to the cloud in real time.

Dashboard & Alert Module
Visualizes system status and generates alerts through a web-based monitoring platform.

IMPLEMENTATION

HARDWARE MODEL

WEB DASHBOARD

CONCLUSION

The SENTINEL system combines IoT sensing, AI-based pest detection, and cloud monitoring for real-time sewer surveillance. By integrating environmental monitoring with intelligent pest classification, it provides a scalable and cost-effective solution for improving sewer safety and maintenance.

FUTURE WORK

- Expand the pest dataset to improve classification accuracy and robustness.
- Incorporate object detection with real-time localization capabilities.
- Add predictive analytics and automated alert prioritization for smarter monitoring.

REFERENCES

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